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Zewail City Project Exhibition
Innovation for Impact

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Rafeeq: AI-Powered Smart Wheelchair Retrofit Kit

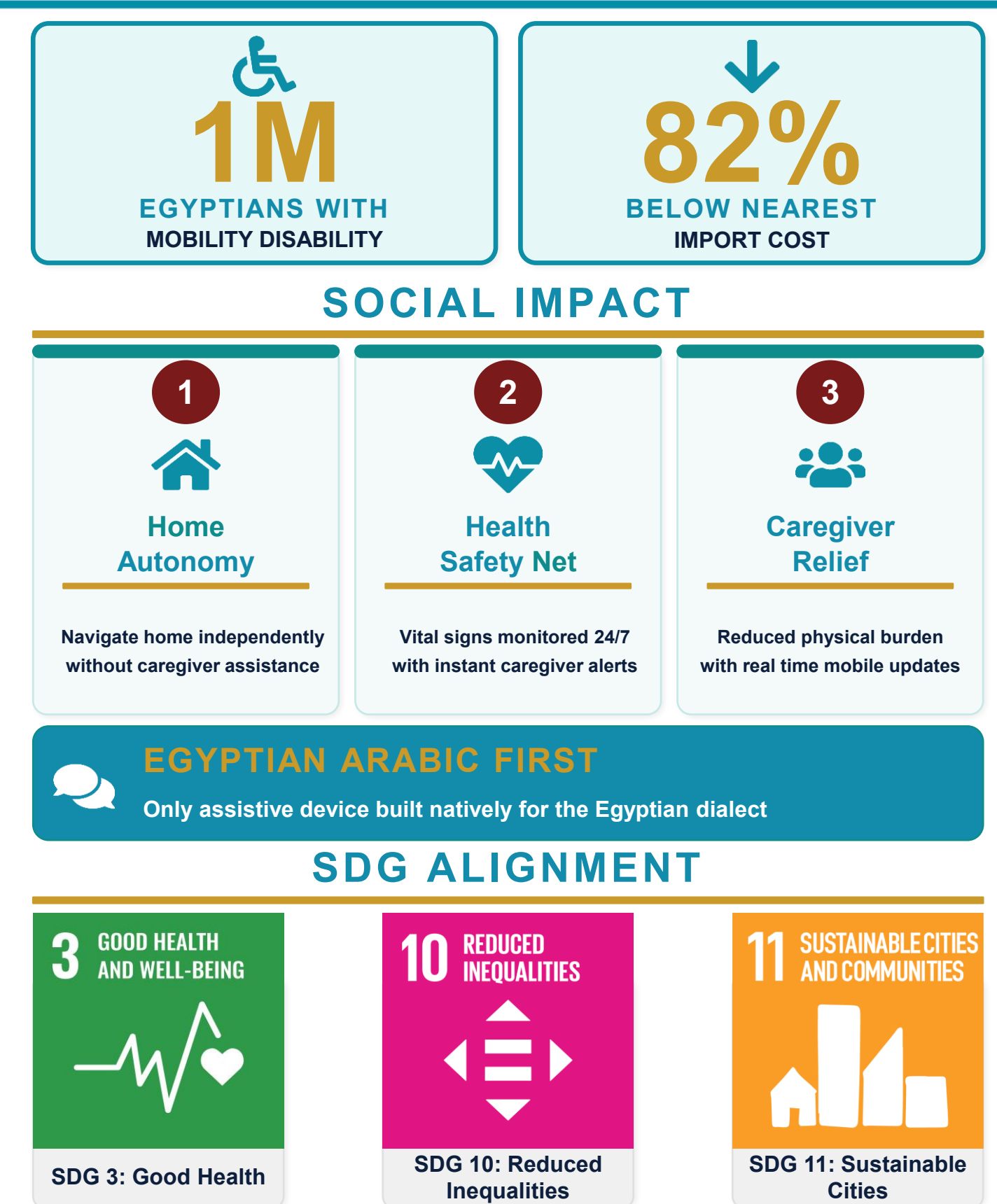
Research Problem

- Egypt has **1 million** wheelchair users **99.35%** have no access to advanced assistive technology
- Smart wheelchairs cost **100,000–480,000 EGP**, up to **20×** the average monthly household income
- No existing solution supports **Egyptian Arabic**, excluding the majority of users
- Family caregivers face severe burden **90.3%** report inadequate income to sustain care

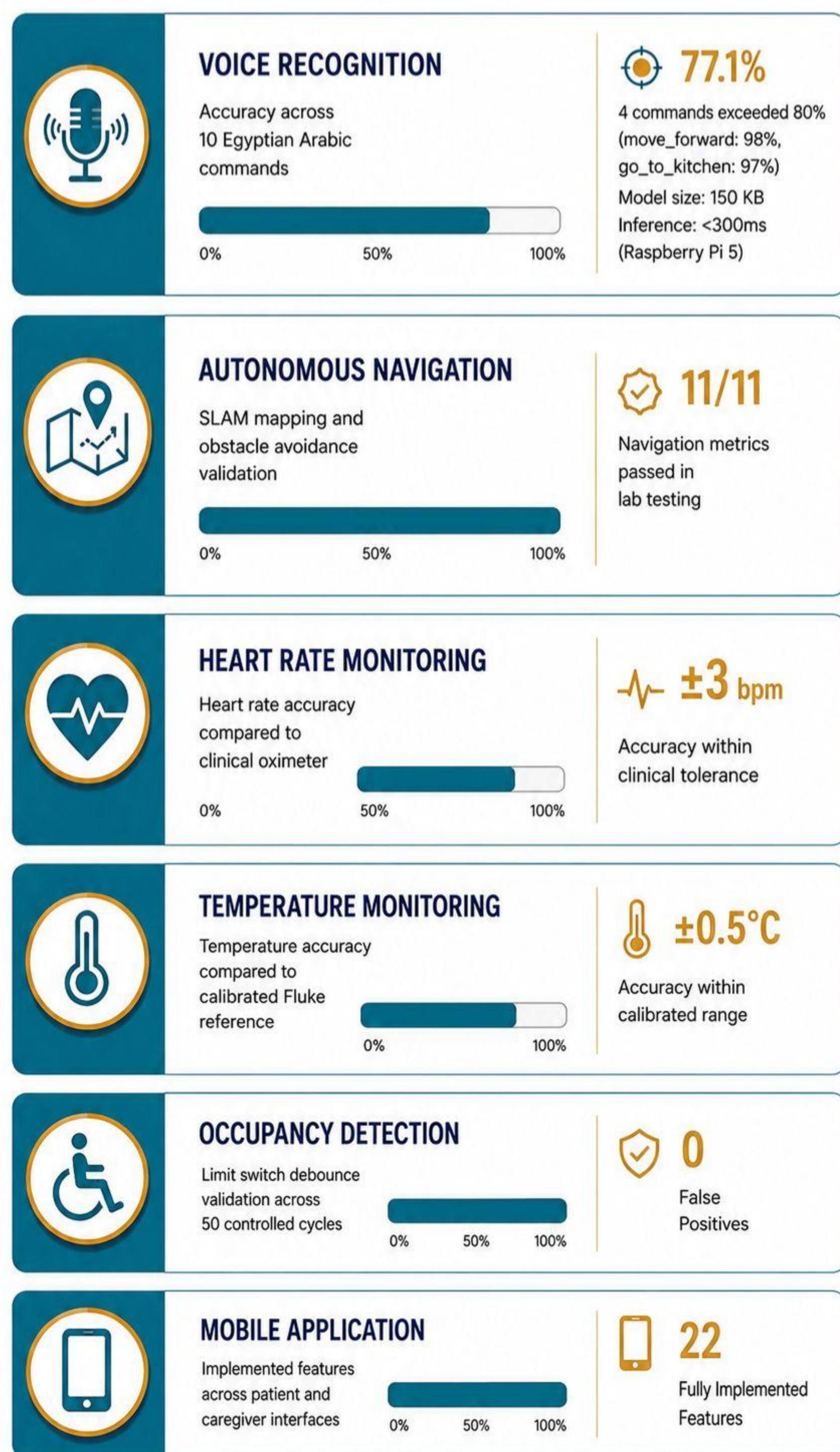
Research methodology

- Documentation-First Agile over **9 months**: design → parallel implementation → integration & testing
- 6 subsystems** developed in parallel and integrated on ROS 2 Jazzy
- Compliance analysis against **ISO 7176 & ISO 13482** standards

Impact / application



Key Findings



Solution/ Prototype

Rafeeq adds intelligence to **any existing wheelchairs** through **6 integrated modules**:

- Add an electrical system and a joystick
- Egyptian Arabic Voice Control
- Autonomous Indoor Navigation
- Mobile application
- AI Assistant in Arabic
- Health Monitoring

Significant Innovation

- 01** First Arabic-Supported Smart Wheelchair
- 02** Offline Speech Recognition
- 03** Universal Retrofit Capability
- 04** Autonomous Indoor Navigation
- 05** Real-Time Health Vitals



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